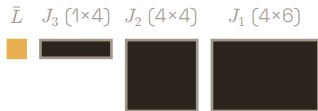
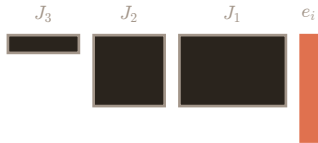


reverse mode: one sweep of vector-Jacobian products



evaluate left to right: every partial product is a row vector
 $1 \times 1 \rightarrow 1 \times 4 \rightarrow 1 \times 4 \rightarrow 1 \times 6 =$ the whole gradient $\nabla_x L$

forward mode: one Jacobian-vector product per input



right to left:
 $6 \times 1 \rightarrow 4 \times 1$
 $\rightarrow 4 \times 1 \rightarrow 1 \times 1$
 $=$ one entry
 $\partial L / \partial x_i$

repeat for $i = 1 \dots 6$: six sweeps